Wireless Voice Transmission and Analysis: Designing a Modulator and Demodulator Prototype

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***Abstract*—This report has the design, implementation, and analysis of a complete wireless communication system that can both transmit and receivie voice signals. The hardware prototype has a modulator circuit operating at a carrier frequency of 75 MHz using a wire antenna for transmission. A corresponding receiver and demodulator circuit extracts the message signal eg the voice signal using a band-pass filter and an audio amplifier. Signal processing and analysis of the transmitted and received audio signals were implemented using in MATLAB with the main focus on the time-domain waveforms and frequency-domain spectral content of an actual voice signal from mic to evaluate the system.**

***Index Terms*—Amplitude Modulation, Demodulation, Carrier Frequency, Signal Processing, LC Tank Circuit, DC Biasing.**

I. Introduction

The fundamental objective of a communication system is the reliable transfer of information from a source to a destination over a medium. In wireless communications, baseband signals—such as human voice, which typically occupies a frequency range of 300 Hz to 3.4 kHz—cannot be efficiently transmitted directly through free space due to the requirement for impractically large antennas and severe attenuation. To overcome this, the process of modulation is employed, wherein the low-frequency baseband signal modifies a high-frequency carrier wave.

This Complex Engineering Problem (CEP) aims to bridge the gap between theoretical communication principles and practical implementation. The task involves designing a robust transmitter (modulator) and receiver (demodulator) prototype to securely transmit audio wirelessly. Furthermore, evaluating the system's performance requires digitizing the received signal via an auxiliary cable to a computer and utilizing graphical user interfaces for spectral and time-domain analysis.

II. System Architecture and Design

This communication module involves an input voice source(a mic), a modulator circuit, wireless channel propagation via wire antennas then a demodulator and finally a data acquisition for digital signal analysis.

A. Modulator and Transmitter Circuit

The transmitter modulates an audio signal form a mic onto the high-frequency carrier. As per the project requirements the carrier frequency must be 5 MHz or higher. For this implementation we have used a frequency of 75 MHz.

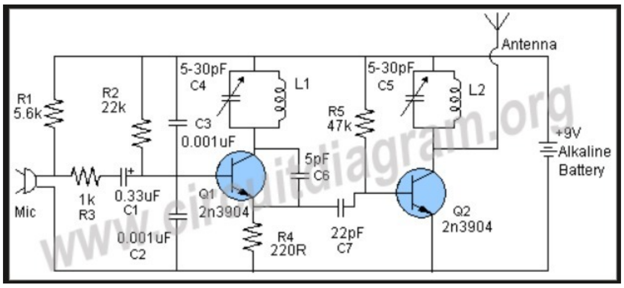

Fig. 1. Schematic diagram of the modulator and transmitter circuit.

As shown in Fig. 1, the circuit has 2N3904 NPN transistors. The first stage acts as a pre-amplifier for the microphone. The modulation basically relies on the LC tank circuit which is made of capacitors (C4, C5) and inductors (L1, L2). The resonant frequency of this LC oscillator dictates the carrier frequency.

B. Demodulator and Receiver Circuit

The main purpose of this receiver is to receive the weak electromagnetic waves transmitted through the wireless medium, which in this case is air, and then tune to the exact carrier wave frequency and retrieve the original message signal.

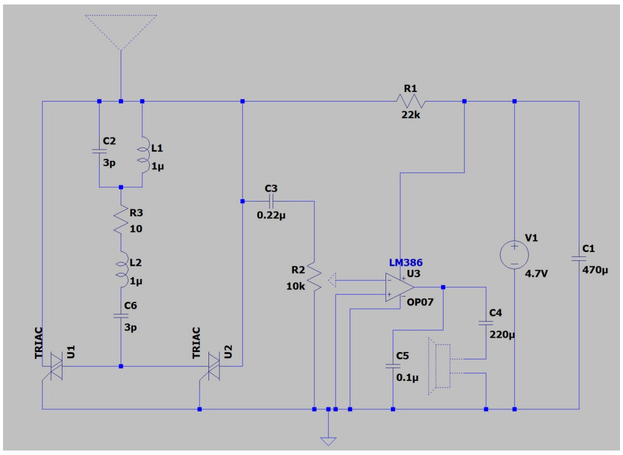

Fig. 2. Schematic diagram of the demodulator and receiver circuit.

The demodulator circuit (Fig. 2) has a LC tuning stage (L1, C2) at the front connected to the receiving antenna. An LM386 op-amp is used as a low-voltage audio amplifier to boost the extracted signal before it is sent to an output speaker and an aux cable for MATLAB for analysis. The capacitor C5 in the circuit diagram functions as a basic low-pass filter to removing the remaining high frequency carrier ripples, leaving the recovered voice envelope.

C. Hardware Prototype Implementation

To validate the theoretical designs, physical prototypes of both the transmitter and receiver were constructed on

breadboards.

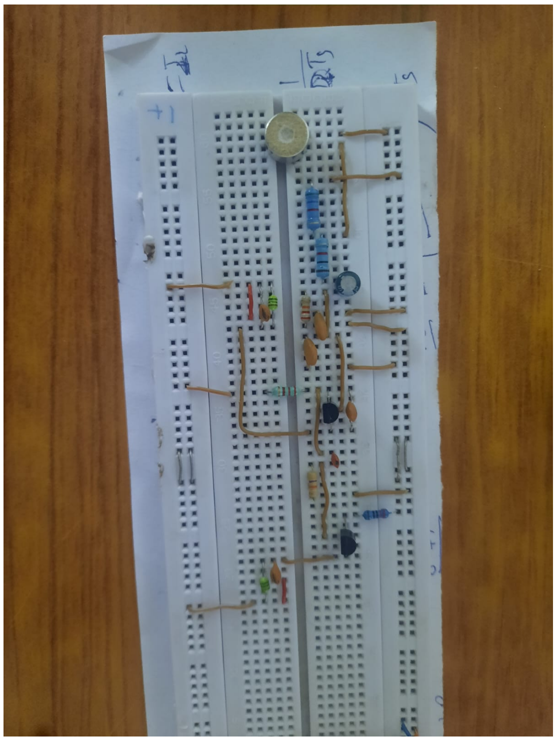

Fig. 3. Physical hardware implementation of the transmitter prototype.

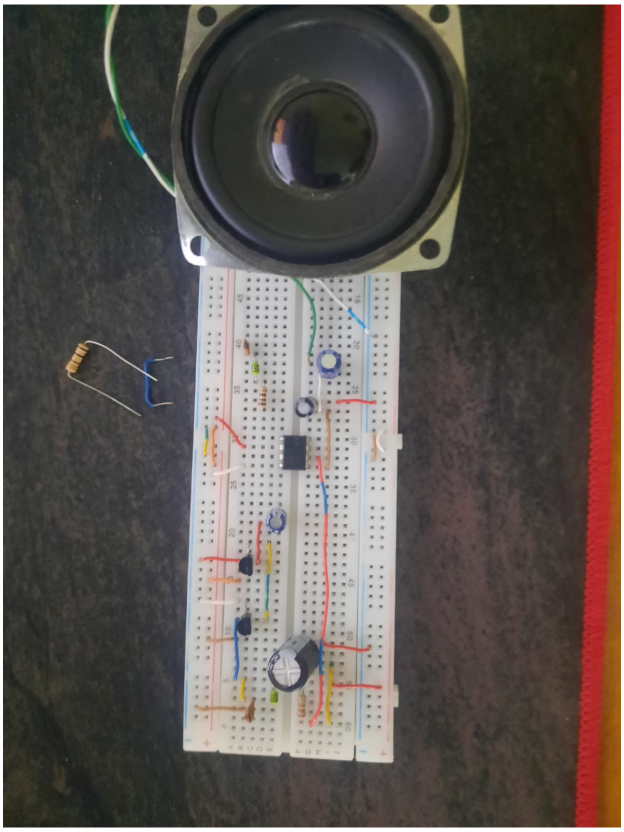

Fig. 4. Physical hardware implementation of the receiver prototype.

The hardware implementation (Fig. 3 and Fig. 4) utilizes manually wound wire antennas for short-range electromagnetic propagation.

III. Mathematical Analysis

A. DC Biasing Analysis

To ensure proper operation of the transmitter, the quiescent operating point (Q-point) for the two 2N3904 NPN transistors (Q_1 and Q_2) is evaluated. A standard forward current gain (β) of 100 and a base-emitter voltage (V_{BE}) of 0.7 V are assumed. For DC analysis, inductors act as short circuits, and capacitors act as open circuits. The supply voltage (V_{CC}) is 9 V.

1) Q_1 (Oscillator/Modulator Stage): Transistor Q_1 is biased using a base resistor $R_2 = 22$ kΩ and an emitter resistor $R_4 = 220\ \Omega$. Applying Kirchhoff's Voltage Law (KVL) to the base-emitter loop:

$$V_{CC} - I_B R_2 - V_{BE} - I_E R_4 = 0 \tag{1}$$

Substituting $I_E = (\beta + 1) I_B$:

$$9 - 0.7 = I_B(22000 + 101 \times 220) \tag{2}$$

$$8.3 = I_B(44220) \implies I_B \approx 187.7\ \mu A\ \mu A\ \mu A_{\mu A(3)}$$

The collector current I_C and collector-emitter voltage V_{CE} are:

$$I_C = \beta I_B = 100 \times 187.7\ \mu A\ \mu A\ \mu A_{\mu A\ =\ 18.77\ \mathrm{mA}(4)}$$

Since the inductor L_1 is a DC short, $V_C = 9$ V. The emitter voltage is $V_E = I_E R_4 \approx I_C R_4 = 4.13$ V.

$$V_{CE} = V_C - V_E = 9 - 4.13 = 4.87 \text{ V} \tag{5}$$

Because $V_{CE} > 0.2$ V, Q_1 is correctly operating in the active region.

2) Q_2 (RF Amplifier Stage): Transistor Q_2 operates as a common-emitter amplifier with a base resistor $R_5 = 47$ kΩ and a grounded emitter ($V_E = 0$ V). Applying KVL to the base loop:

$$V_{CC} - I_B R_5 - V_{BE} = 0 \tag{6}$$

$$9 - 0.7 = I_B(47000) \implies I_B \approx 176.6\ \mu A\ \mu A\ \mu A_{\mu A(7)}$$

The collector current is:

$$I_C = \beta I_B = 100 \times 176.6\ \mu A\ \mu A\ \mu A_{\mu A\ =\ 17.66\ \mathrm{mA}(8)}$$

With the inductor L_2 acting as a DC short, the collector voltage is $V_C = 9$ V. Thus:

$$V_{CE} = V_C - V_E = 9 - 0 = 9 \text{ V} \tag{9}$$

Q_2 is also safely operating in the active region, allowing for maximum AC voltage swing across the LC tank.

B. Carrier Frequency Calculation

The transmission frequency depends on the resonant frequency of the tuned LC circuit. The theoretical resonant frequency f_c is given by:

$$f_c = \frac{1}{2\pi\sqrt{LC}} \tag{10}$$

The target transmission frequency is $f_c = 75$ MHz. To achieve this, we configure the capacitor C_4 and inductor L_1. Selecting a standard inductor value of $L_1 = 1.5uH$

C. Low-Pass Filter Cutoff

For the demodulator, a Low-Pass Filter (LPF) is used to reject the 75 MHz carrier and pass the voice signal.

For standard voice signals, the components are chosen such that f_{cutoff} is lower than carrier.

IV. Signal Processing and Analysis

The received signal was divided into two paths: one driving the speaker and the other interfaced with a laptop via an AUX cable for deep analysis using MATLAB.

A. Input Signal Characteristics

Fig. 5 demonstrates the time and frequency domains of the generated input audio, "Hello, This voice is generated by MATLAB."

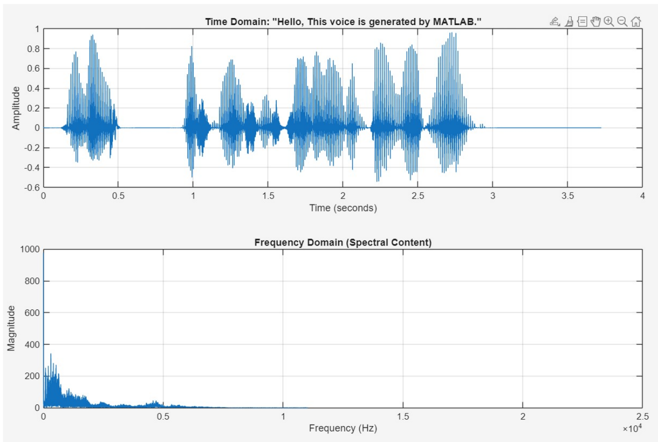

Fig. 5. Time and Frequency domain plots of the original input voice signal.

The time-domain plot illustrates clean, high-amplitude voltage variations representing the spoken words. The frequency domain shows the spectral content heavily concentrated at the lower end of the spectrum typical for human speech.

B. Output Signal Characteristics

Following modulation, wireless transmission, and demodulation, the received signal was captured and plotted (Fig. 6).

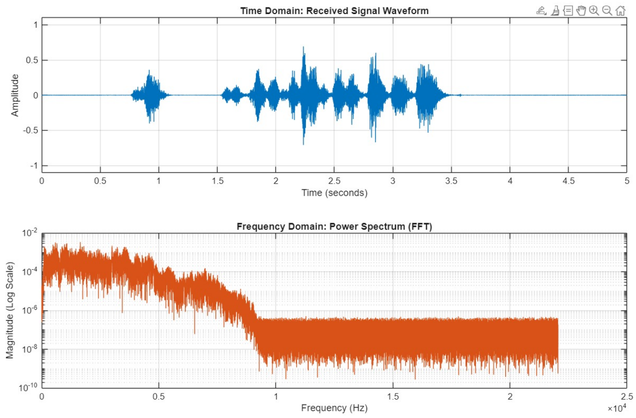

Fig. 6. Time and Frequency (FFT) domain plots of the received signal.

The time-domain waveform of the received signal retains the fundamental envelope of the input, indicating successful transmission. However, there is a visible reduction in amplitude and the introduction of channel noise. The Fast Fourier Transform (FFT) power spectrum highlights a dense magnitude clustering at baseband frequencies. The noise floor is elevated compared to the ideal input, which is a standard consequence of analog wireless channels and hardware-induced harmonic distortions.

V. Conclusion

This project achieved the design of a wireless communication system from beginning to end. The modulator was designed a way that it could generate a certain carrier frequency of 75 MHz through an LC tank circuit that was adjusted. The demodulator detected the voice envelope and LM386 audio amplifier. The advanced digital signal analysis done through MATLAB proved that both the transmission and reception of signal was valid because the spectrum was very much intact.

References

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